3D-2D analysis of aircraft

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Problem statement

Assumptions :

1) Internal parameters of the camera are known (f, FOV, #pixels)
2) Pinhole perspective camera with very small FOV, symmetric, callibrated, without distortions
3) Target is at very high distance (> 1 km) relative to its size and FOV is very small
4) Aircraft is symmetric and represented as cross (two points on body, two points on edges of wings)
5) Distance to center of body is known, D
6) Distance to center of wings is same, D (not true but we will assume)
7) Velocity vector represents vector of body and known and not noisy
8) Coordinates of 4 keypoints are known and not noisy

In short, we will assume that we live in the perfect world, but in some cases we will prefer Least Squares solution to allow for noise. Because if we go for an exact solution we will probably end up without any solution.

Given:

Camera *f, H_pixel, W_pixel*. See appendix how to compute *f*

$(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$ pixel coordinates of body edges and tail edges. We will not use them since we are going to use only **normalized** coordinates. See appendix.

$(u_1, v_1), (u_2, v_2)$, - **normalized** coordinates of body edges in image

$(u_3, v_3), (u_4, v_4)$ - **normalized** coordinates of wings edges in image

From now on , all 2D coordinates in image plane in this document are normalized.

D - distance to the aircraft from the focal mode of camera (center of body, center of wings)

$\overline{V} = (V_X, V_Y, V_z)$ - velocity vector of aircraft (unit vector or velocity in meters/sec)

We want to compute:

3D locations of all 4 points:

Edges of body :

$\overline{P_1} = (X_1, Y_1, Z_1), \overline{P_2} = (X_2, Y_2, Z_2),$

Edges of wings :

$\overline{P_3} = (X_3, Y_3, Z_3), \overline{P_4} = (X_4, Y_4, Z_4),$

From these locations we will easily have 6DOF of the aircraft and its sizes. See appendix

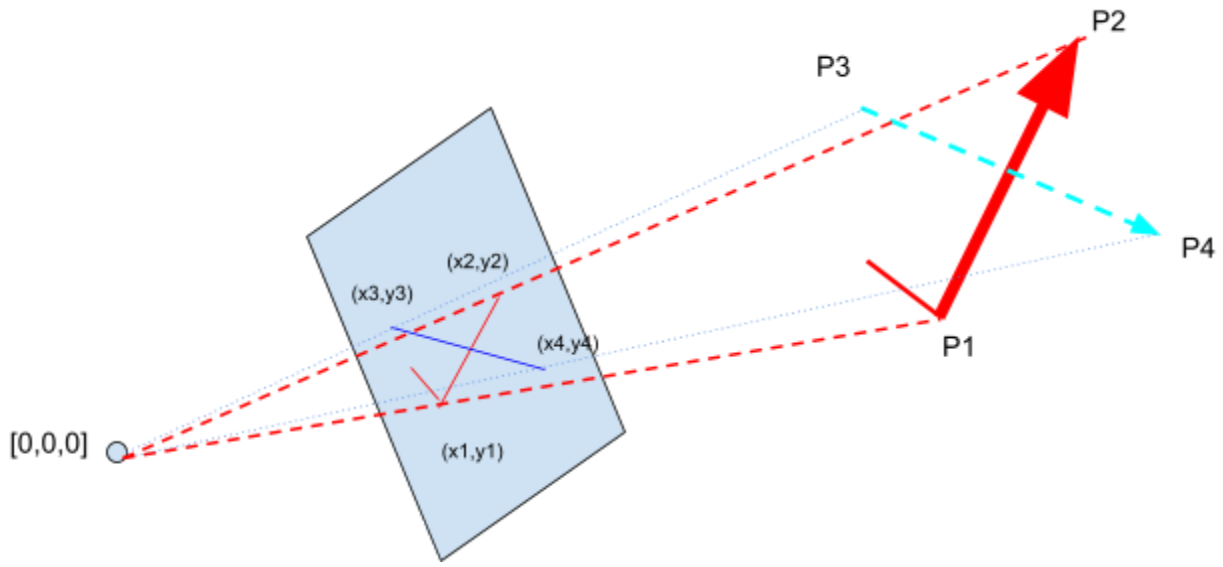


Figure 1: Schematic representation of 3D aircraft and it edges coordinates and their projection on virtual image plane (the real image plane is from other side of (0,0,0))

Body of aircraft:

If required, read some theory in Appendix .

We will present here two solutions, first one uses body vector direction and next - the range. Second, first uses range and then the vector direction.

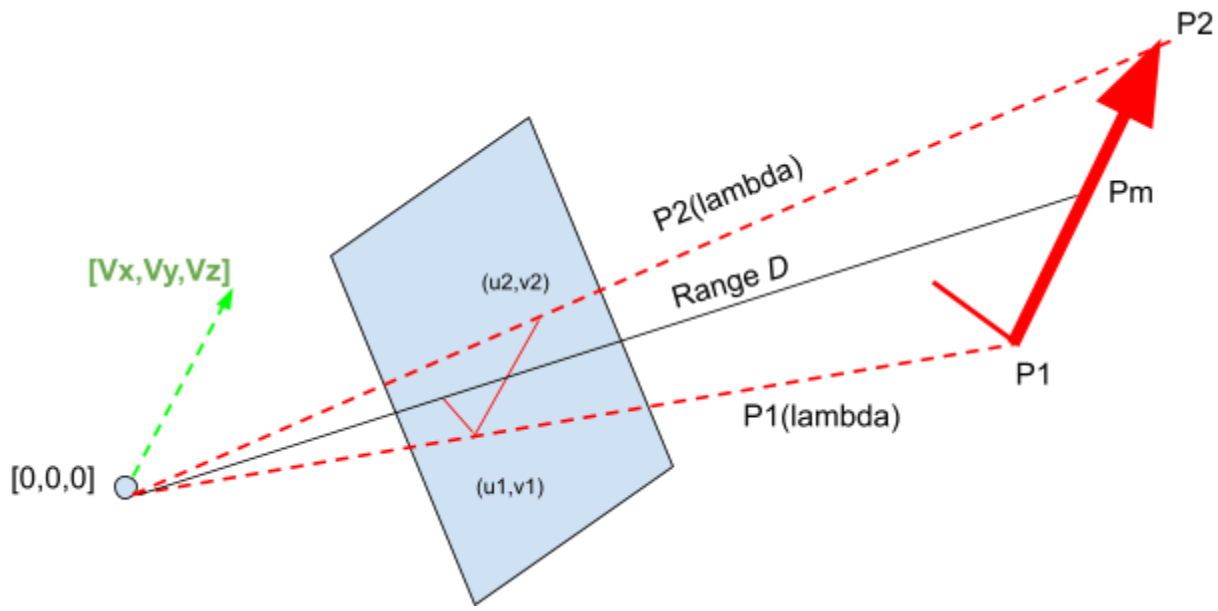


Figure 2: Schematic illustration. (u_i, v_i) - coordinate in 2D; $\vec{P}_i = (X_i, Y_i, Z_i)$ coordinate in 3D; $P_i(\lambda)$ - ray corresponding to pixel, $\vec{Pm}$ - mid-point, $\vec{V} = [V_x, V_Y, V_Z]$ - velocity vector

Solution #1

The vector $\vec{M} = \vec{P}_2 - \vec{P}_1$ is parallel to $\vec{V} = (V_X, V_Y, V_Z)$

So $\vec{M} = s\vec{V}$, for some scalar *s*

Then,

$$\lambda_2 u_2 - \lambda_1 u_1 = sV_X$$
$$\lambda_2 v_2 - \lambda_1 v_1 = sV_Y$$
$$\lambda_2 - \lambda_1 = sV_Z$$

Let's $\tilde{\lambda}_1 = \lambda_1/s$, $\tilde{\lambda}_2 = \lambda_2/s$

$$\begin{bmatrix} u_2 & -u_1 \\ v_2 & -v_1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \tilde{\lambda}_2 \\ \tilde{\lambda}_1 \end{bmatrix} = \begin{bmatrix} V_X \\ V_Y \\ V_Z \end{bmatrix}$$

Use LSE solution:

$$A = \begin{bmatrix} u_2 & -u_1 \\ v_2 & -v_1 \\ 1 & -1 \end{bmatrix}; b = \begin{bmatrix} V_X \\ V_Y \\ V_Z \end{bmatrix}$$

$$\begin{bmatrix} \tilde{\lambda}_2 \\ \tilde{\lambda}_1 \end{bmatrix} = (A^T A)^{-1} A^T b$$

We still need to find *s,* we will use range *D*

Coordinate of the middle point is

$$0.5\left(\lambda_1 \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix}\right) = 0.5s\left(\tilde{\lambda}_1 \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} + \tilde{\lambda}_2 \begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix}\right) = 0.5s\vec{a}$$

Use range

$0.5s||a||$=D $\Rightarrow$ $s = 2D/||a||$

$s\tilde{\lambda_1} = \lambda_1$, $s\tilde{\lambda_2} = \lambda_2$

Solution #2

Start with the middle point in 2D

$u_m = 0.5(u_1 + u_2)$

$v_m = 0.5(v_1 + v_2)$

Remark: we can also use mid-body-point from the keypoint detector here.

Middle point in 3D

$\vec{P}_m = \lambda_m (u_m, v_m, 1)^T$

$\sqrt{\lambda^2_m u_m^2 + \lambda^2_m v_m^2 + \lambda^2_m} = D \quad \Rightarrow \lambda_m = \frac{D}{\sqrt{u_m^2 + v_m^2 + 1}}$

Substitute λ_m and we get (X_m, Y_m, Z_m)

Now we want to build a 3D segment with given 3D middle point and direction like $\vec{V}$ which extends symmetrically to both sides by unknown amount h until it meets rays P_1 and P_2. The meeting points are $\overline{P}_1$ and $\overline{P}_2$.

$\vec{P}_1 = \vec{P}_m + \vec{V}h$ and $\vec{P}_2 = \vec{P}_m - \vec{V}h$

Then we get 6 equations + 3 unknowns (overcomplete) :

$X_m + V_X h = \lambda_1 u_1$

$Y_m + V_Y h = \lambda_1 v_1$

$Z_m + V_Z h = \lambda_1$

$X_m - V_X h = \lambda_2 u_2$

$Y_m - V_Y h = \lambda_2 v_2$

$Z_m - V_Z h = \lambda_2$

Converted to matrix form :

$$\begin{bmatrix} u_1 & 0 & -V_X \\ v_1 & 0 & -V_Y \\ 1 & 0 & -V_Z \\ 0 & u_2 & V_X \\ 0 & v_2 & V_Y \\ 0 & 1 & V_Z \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ h \end{bmatrix} = \begin{bmatrix} X_m \\ Y_m \\ Z_m \\ X_m \\ Y_m \\ Z_M \end{bmatrix}$$

Again

$$A = \begin{bmatrix} u_1 & 0 & -V_X \\ v_1 & 0 & -V_Y \\ 1 & 0 & -V_Z \\ 0 & u_2 & V_X \\ 0 & v_2 & V_Y \\ 0 & 1 & V_Z \end{bmatrix}; b = \begin{bmatrix} X_m \\ Y_m \\ Z_m \\ X_m \\ Y_m \\ Z_M \end{bmatrix}$$

Use LSE:

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ h \end{bmatrix} = [A^T A]^{-1} A^T b$$

Wings of aircraft:

Now we will take care of the wings. We suppose that we know direction of the body $\vec{V}$ and distance D to the midpoint between wings.

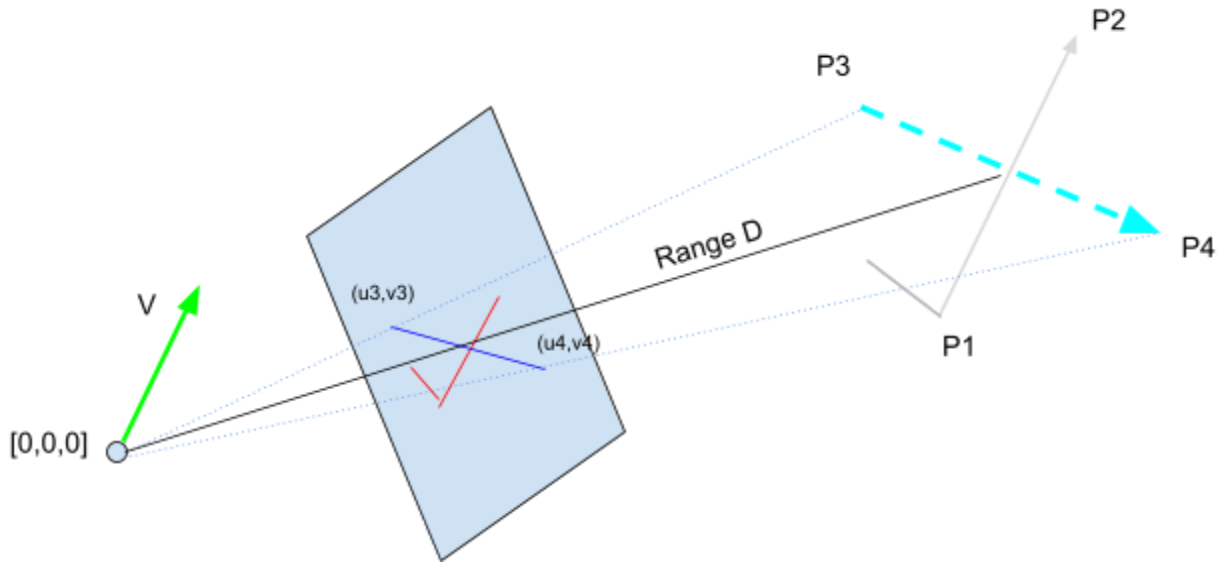


Figure 3: Similar to previous figure, but this time we are looking for segment which is perpendicular to $\vec{V}$

We want to find wing edges in 3D on corresponding rays.

$P_3 = \lambda_3 \vec{d}_3 = \lambda_3 (u_3, v_3, 1)^T$

$$P_4 = \lambda_4 \vec{d}_4 \;=\; \lambda_4 (u_4,\, v_4,\, 1)^T$$

Use the distance *D* to the center of mass:

$$\left\| 0.5\left(\lambda_3 \vec{d}_3 \;+\; \lambda_4 \vec{d}_4\right) \right\|_2^2 = D^2$$

(*) $\lambda_3{}^2(\vec{d}_3 \cdot \vec{d}_3) + \lambda_4{}^2(\vec{d}_4 \cdot \vec{d}_4) + 2\lambda_3\lambda_4\;(\vec{d}_3 \cdot \vec{d}_4) \;=\; 4D^2$, where

$\vec{d}_3 \cdot \vec{d}_3 = u_3{}^2 + v_3{}^2 + 1$

$\vec{d}_4 \cdot \vec{d}_4 = u_4{}^2 + v_4{}^2 + 1$

$\vec{d}_3 \cdot \vec{d}_4 = u_3 u_4 + v_3 v_4 + 1$

The vector of wings direction is $\vec{P}_3 - \vec{P}_4 \;\perp\; \vec{V} \quad \Rightarrow \quad \left(\vec{P}_3 - \vec{P}_4\right) \cdot \vec{V} = 0$

$\lambda_3 \vec{d}_3 \cdot \vec{V} \;- \lambda_4 \vec{d}_4 \cdot \vec{V} = 0$

$\lambda_4 \;=\; \lambda_3 \frac{\vec{d}_3 \cdot \vec{V}}{\vec{d}_4 \cdot \vec{V}} = \lambda_3 \frac{\left(u_3 V_X + v_3 V_Y + V_Z\right)}{\left(u_4 V_X + v_4 V_Y + V_Z\right)} = \lambda_3 a$

Let's substitute it to **(*)** and we will get one equation with one unknown

$$\lambda_3{}^2(\vec{d}_3 \cdot \vec{d}_3) + \lambda_3{}^2 a^2(\vec{d}_4 \cdot \vec{d}_4) + 2a\lambda_3{}^2\;(\vec{d}_3 \cdot \vec{d}_4) \;=\; 4D^2$$

$$\lambda_3{}^2\left[(\vec{d}_3 \cdot \vec{d}_3) + a^2(\vec{d}_4 \cdot \vec{d}_4) + 2a\;(\vec{d}_3 \cdot \vec{d}_4)\right] \;=\; 4D^2$$

Eventually,

$$\lambda_3 = \frac{2D}{\sqrt{\left[(\vec{d}_3 \cdot \vec{d}_3) + a^2(\vec{d}_4 \cdot \vec{d}_4) + 2a\;(\vec{d}_3 \cdot \vec{d}_4)\right]}}$$

And after we get λ_3 **,** we compute $\lambda_4 = \lambda_3 a$

Remark: same way we can compute coordinates of any Left/Right pair of keypoints.

What is next? Now we have 3D coordinates of 4 edges of the aircraft we can :

1) Compute 6DOF
2) Compute body length and wings-span
3) Predict and filter

6 DOF of aircraft:

We will suppose that the local coordinate system of the aircraft is as follows:

a) [0,0,0] is center of the body
b) Tail to head points in Y direction
c) Left to right wing is X direction
d) Z is up (like direction of the body of pilot)

Compute translation:

The body center is $P_m = 0.5(P_1 + P_2)$ and this is our **t_vec** (translation of local frame into world frame): T= t_vec = P_m

Compute axes in world coordinates

$$y = \frac{P_1 - P_0}{||P_1 - P_0||}$$

$$x = \frac{P_4 - P_3}{||P_4 - P_3||}$$

$$z = \frac{x \times y}{||x \times y||}$$

Rotation matrix :

$$R = \begin{bmatrix} | & | & | \\ \mathbf{x} & \mathbf{y} & \mathbf{z} \\ | & | & | \end{bmatrix}$$

It can be converted to r_vec
r_vec = Rodrigues(R)

From rotation matrix R:

$$R = \begin{bmatrix} r_{00} & r_{01} & r_{02} \\ r_{10} & r_{11} & r_{12} \\ r_{20} & r_{21} & r_{22} \end{bmatrix}$$

$$\text{pitch} = \arcsin(-r_{20})$$

$$\text{yaw} = \arctan 2(r_{10}, r_{00})$$

$$\text{roll} = \arctan 2(r_{21}, r_{22})$$

Appendix

Normalized coordinates
(u, v) = ((x_pixel - cx)/f, (y_pixel - cy)/f)) , where cx = W_pixel/ 2, cy = H_pixel/2

Focal length of camera with FOV, and W and pixel size S is
$f = \frac{WS}{2\,tan(FOV/2)}$, in our case *f* is in pixels and not mm, so *S= 1*

Perspective camera
Each pixel (u,v) in a primitive perspective (pin-hole) camera is represented by a 3D ray in space. The ray starts at the optical center of the camera and represents all the points in space along the ray which can be projected into the corresponding pixel. The optical center of the camera is [0,0,0], the center of the virtual image plane is located in [0,0,f].

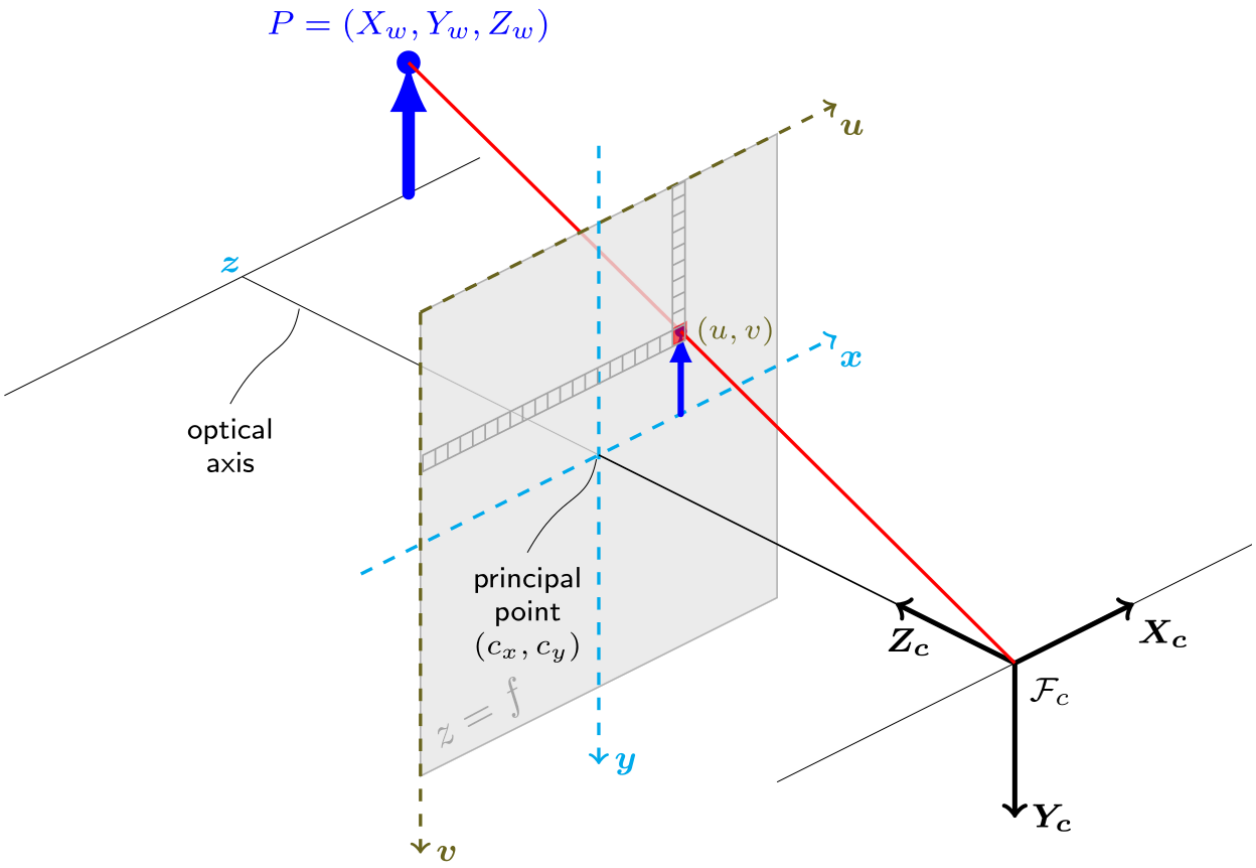


Figure 2: pinhole camera. TODO: **In our notation x,y is u,v. Sorry don't have time to change the figure**

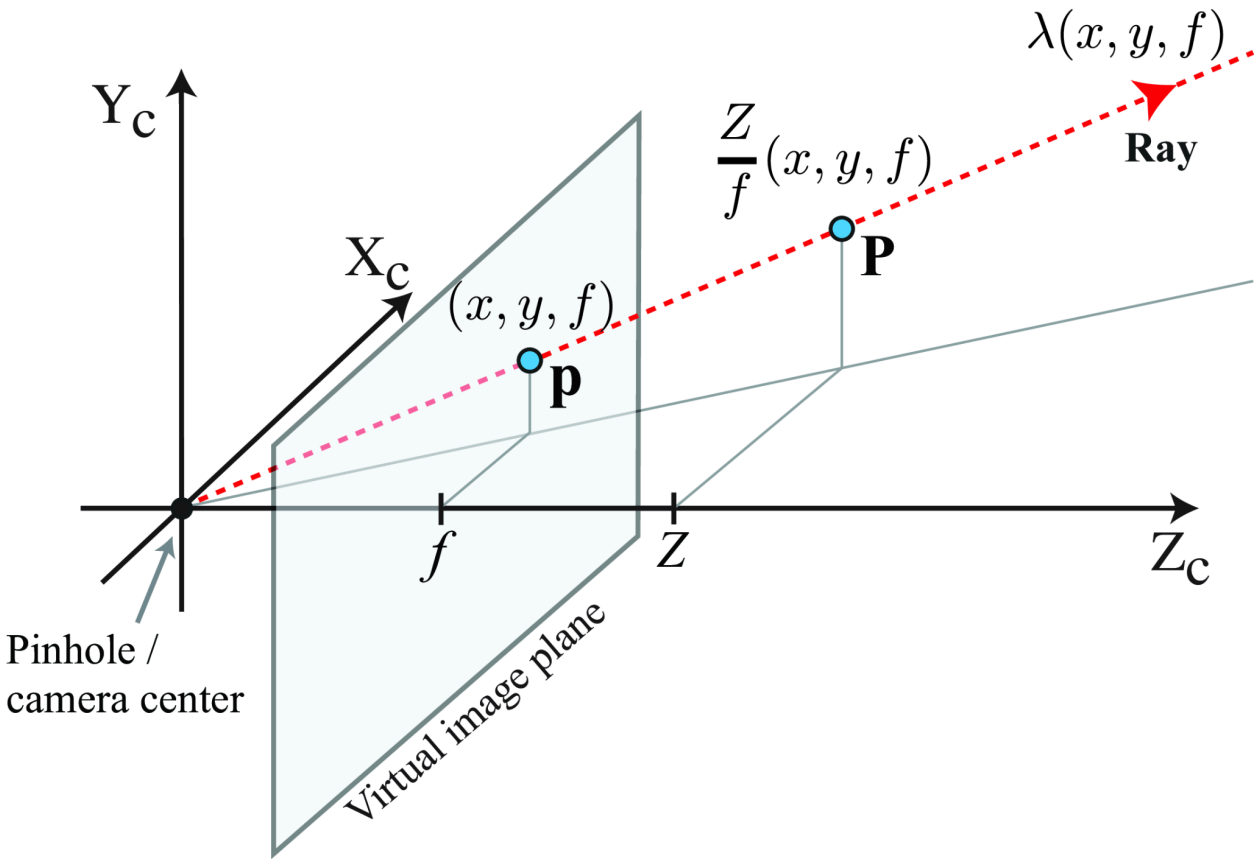


Figure 3: Correspondence between rays and pixels

We we will represent each ray as
$P = \lambda\overline{d} = \lambda\,[u\,v\,1]$, in fact we just need to solve for λ

Imagine that [u v 1] is just a direction vector and λ is just scaling to "send" our point along the ray in space.

Remark: since FOV is very small and distance is very large (see Assumptions), the rays near the target are approximately parallel. It means that the projected object behaves almost the same as in orthographic projection. For example the middle of a segment in 3D is projected to the middle of the segment in 2D.

Latex equations:

\begin{bmatrix} x_2 & -x_1\\ y_2 & -y_1\\ 1 & -1 \end{bmatrix} \begin{bmatrix} \tilde\lambda_2 \\ \tilde\lambda_1\\ \end{bmatrix} = \begin{bmatrix} V_X \\ V_Y\\ V_Z \end{bmatrix}

$$0.5\left(\lambda_1\begin{bmatrix} x_1 \\ y_1\\ 1 \end{bmatrix} + \lambda_2\begin{bmatrix} x_2 \\ y_2\\ 1 \end{bmatrix} \right) =0.5s \left(\tilde\lambda_1\begin{bmatrix} x_1 \\ y_1\\ 1 \end{bmatrix} + \tilde\lambda_2\begin{bmatrix} x_2 \\ y_2\\ 1 \end{bmatrix} \right)=0.5s\vec{a}$$

$$.5\left(\lambda_1\begin{bmatrix} u_1 \\ v_1\\ 1 \end{bmatrix} + \lambda_2\begin{bmatrix} u_2 \\ v_2\\ 1 \end{bmatrix} \right) =0.5s \left(\tilde\lambda_1\begin{bmatrix} u_1 \\ v_1\\ 1 \end{bmatrix} + \tilde\lambda_2\begin{bmatrix} u_2 \\ v_2\\ 1 \end{bmatrix} \right)=0.5s\vec{a}$$

$$\begin{bmatrix} u_1 & 0 & -V_X \\ v_1 & 0 & -V_Y\\ 1 & 0 & -V_Z \\ 0 & u_2 & V_X \\ 0 & v_2 & V_Y\\ 0 & 1 & V_Z \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2\\h \end{bmatrix} = \begin{bmatrix} X_m \\ Y_m \\ Z_m \\ X_m \\ Y_m \\ Z_M \end{bmatrix}$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\h \end{bmatrix} = [A^TA]^{-1}A^T b$$